

ML-Assisted Optimization for FePt granular films

Yukiko K. Takahashi^{1,2}, *Senior Member, IEEE*

¹Natinoal Institute for Materials Science, Tsukuba, 305-0047, Japan, takahashi.yukiko@nims.go.jp

²RIEC, Tohoku University, Sendai, 980-8577, Japan

We present a data-driven framework for the rapid optimization of FePt–BN–C granular media for next-generation magnetic recording. By integrating automated data extraction with multi-objective active learning, we efficiently explored a high-dimensional processing space and identified media with grain sizes of ~5 nm and coercivity exceeding 2 T within only two optimization cycles. The results reveal a synergistic mechanism: carbon enhances nucleation and grain refinement, while BN forms structured intergranular phases that improve magnetic isolation. This approach demonstrates a general strategy for accelerating materials discovery beyond conventional trial-and-error methods.

Index Terms—FePt granular film, machine learning, microstructure.

I. INTRODUCTION

THE rapid expansion of artificial intelligence, cloud computing, and data-centric technologies has significantly increased the demand for high-density data storage. Hard disk drives (HDDs) remain a key technology due to their scalability and cost efficiency; however, further increases in areal recording density require substantial advances in magnetic recording media. In particular, achieving densities beyond 3 Tbit/in² necessitates reducing the grain size to below approximately 5 nm while maintaining sufficient thermal stability and writability.

L1₀-ordered FePt has been widely recognized as a promising material for next-generation recording media because of its extremely high magnetic anisotropy. Nevertheless, the simultaneous realization of ultra-fine grain size and high coercivity remains a fundamental challenge, as grain refinement typically leads to a reduction in magnetic anisotropy and coercivity. Furthermore, the microstructure and magnetic properties of FePt granular films are highly sensitive to sputtering conditions and segregant composition, resulting in a complex and high-dimensional parameter space that is difficult to optimize using conventional trial-and-error approaches.

To address this challenge, we developed a data-driven, closed-loop optimization framework that integrates automated data analysis with machine-learning-based experimental design. In this study, we apply this approach to FePt–BN–C granular media, where boron nitride (BN) and carbon (C) are used as mixed segregants. By combining the complementary roles of these materials and leveraging data-driven optimization, we demonstrate rapid discovery of media exhibiting both sub-5 nm grain size and high coercivity.

II. EXPERIMENTAL METHOD

Thin films of FePt–C, FePt–BN, and FePt–BN–C were prepared using an ultrahigh vacuum co-sputtering system. The base pressure was on the order of 10^{−7} Pa, and the film thickness was fixed at 6 nm for all samples. The deposition temperature was optimized depending on the composition, typically in the range of 600–700 °C to promote L1₀ ordering. A thin protective

carbon layer was deposited on top of the films to prevent surface degradation.

A closed-loop optimization workflow was implemented by integrating experimental measurements with a data-driven platform. Structural and magnetic characterization data, including transmission electron microscopy (TEM) images, X-ray diffraction (XRD) patterns, and magnetization curves, were uploaded to the Research Data Express (RDE) system. Within this platform, microstructural parameters such as grain size were automatically extracted from TEM images using a deep-learning-based image analysis model [1], while magnetic properties such as coercivity were obtained from processed magnetization curves.

The extracted parameters were used as objective functions in a multi-objective optimization scheme. We employed the Probability of Target Range (PTR) algorithm [2] implemented in the NIMO software package [3] to propose new experimental conditions. The optimization targeted grain sizes below 6 nm and coercivity above 2 T. Based on the predicted outcomes, new sputtering conditions were selected, and the corresponding samples were synthesized and characterized. This cycle of data acquisition, analysis, and machine-learning-guided proposal was repeated to efficiently explore the parameter space.

III. RESULTS AND DISCUSSION

The data-driven optimization rapidly converged toward the target property region within only two experimental cycles. Starting from initial datasets consisting of FePt–C and FePt–BN films, the first optimization cycle yielded FePt–BN–C films with a grain size of approximately 5.3 nm and a coercivity exceeding 3 T. In the second cycle, further refinement led to a grain size of 4.9 nm while maintaining coercivity above 2 T. These results demonstrate that the closed-loop framework enables efficient exploration of a complex parameter space with a minimal number of experiments.

Microstructural analysis revealed that the mixed use of BN and C plays a crucial role in achieving this performance. Carbon promotes a high nucleation density during film growth, which effectively suppresses lateral grain growth and leads to grain refinement. In contrast, BN forms partially crystalline, layered intergranular phases that enhance magnetic isolation between

neighboring grains. The coexistence of these two mechanisms enables independent control of grain size and magnetic isolation, thereby mitigating the conventional trade-off between grain refinement and coercivity.

An important finding of this study is that the coercivity enhancement in BN-containing samples cannot be explained solely by the apparent degree of L1₀ ordering estimated from X-ray diffraction. Although the order parameter evaluated from the (001)/(002) intensity ratio appears lower than that of FePt–C films, this reduction is likely influenced by the presence of in-plane variants, which lead to an underestimation of the true ordering. Taking this into account, the intrinsic anisotropy of FePt grains is expected to be comparable among the samples. Therefore, the significantly higher coercivity observed in BN-containing films indicates that intergranular boundary structure plays a dominant role in determining magnetic properties. The layered BN-containing boundaries are likely to suppress the formation of sub-nanometer magnetic bridges between grains, resulting in more uniform magnetic isolation and enhanced coercivity.

These results highlight that intergranular phase engineering is a key design parameter for next-generation magnetic recording media. The combination of BN and C enables a tunable balance between nucleation-driven grain refinement and boundary-controlled magnetic isolation. More broadly, the demonstrated data-driven framework provides a powerful approach for accelerating materials discovery in systems characterized by complex, multi-parameter dependencies.

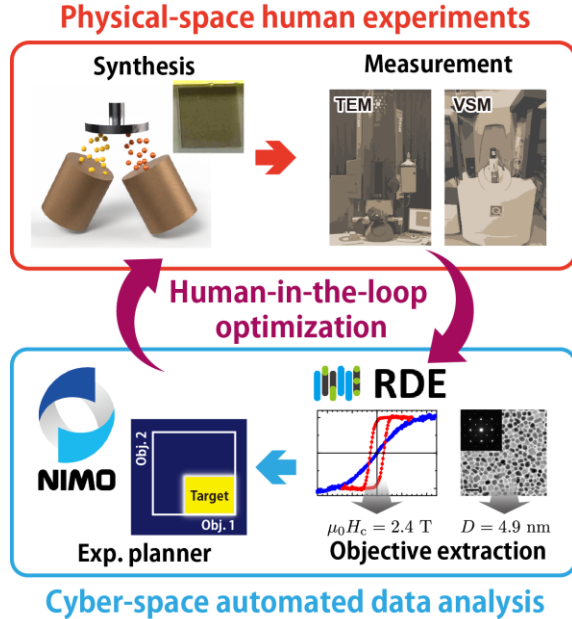


Fig. 1 Schematic illustration of the closed-loop optimization framework implemented in the Research Data Express (RDE) system [4]. The workflow integrates sputter deposition, automated data accumulation, and characterization (TEM, XRD, and VSM) with machine-learning-guided condition selection using the PTR algorithm (NIMO), enabling data-efficient optimization³

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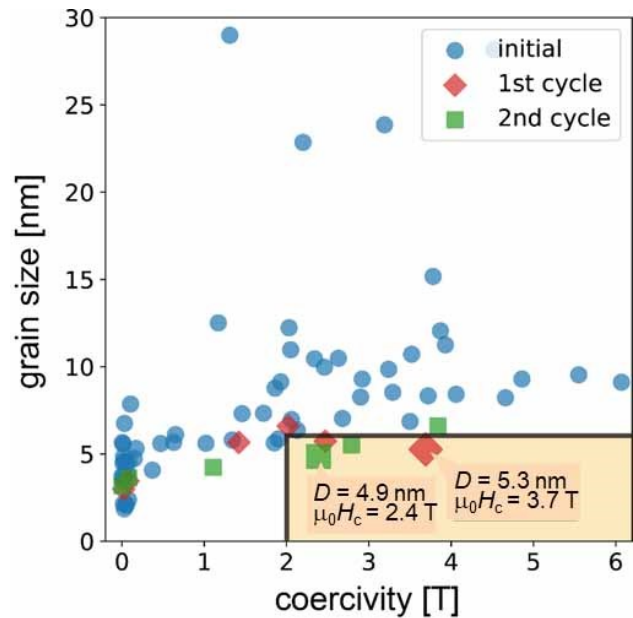


Fig. 2 Machine-learning-guided optimization of grain size and coercivity. Transition of average grain size D and coercivity $\mu_0 H_c$ as a function of optimization cycle. Initial datasets for FePt–C and FePt–BN films were followed by two cycles of PTR-based proposals, synthesis, and automated data analysis within RDE. The yellow shaded region indicates the target window ($D \leq 6 \text{ nm}$ and $\mu_0 H_c \geq 2 \text{ T}$). FePt–BN–C films identified in the first and second cycles satisfy both criteria, demonstrating efficient convergence toward the predefined target region.